

THINGS TO KNOW:

1. Lab report must contain following sections: (order must be maintained)
 - a) Title /Question
 - b) Theory: The brief overview of the concept /techniques/syntax/idea/tools etc used
 - c) Steps and Commands: The steps performed along with involved commands
 - d) Output: Screenshot of the output
2. Output screen should be captured (use snipping tool), printed and attached in the report.
3. The output screen should reflect the unique identity of each students for that include the following things in commands (to print message in screen) and/or names and/ or paths :
 - Personal Information (Name, Roll No. and Section)
4. Contents should be written on single side of A4 sized paper.
5. Cover page and contents page should be attached in the report appropriately.
6. The works must be submitted within specified deadline.

Content Page Format

Subject: Database Administration

List of Lab Works

Lab No.	Title /Question	Submission Date	Signature
1(a)	This is sample title	2079/03/15	
1(b)	This is another title	2079/03/17	

Lab works / Part-3 / DBA/ BIT-6th semester

First Create PL/SQL stored procedure that displays the information about lab no. and your personal information in required places (CDB and PDB) and execute the stored procedure at beginning or end of the each lab task (for those lab works where SQL Plus is used)

Sample PL/SQL stored procedure:

```
CREATE OR REPLACE PROCEDURE my_information IS
BEGIN
    DBMS_OUTPUT.PUT_LINE('Name: Ram Prasad Sharma/Roll No.: 102 / Section: A');
END my_information;
/
```

Execution:

```
SET SERVEROUTPUT ON;
EXEC my_information;
```

11. Working with RMAN backup and Recovery

Perform following tasks to demonstrate how backup and recovery is carried out using RMAN

(List the steps involved with commands and include the screenshots of successful operations)

- a) Preparation and Configuration for backup
 - Ensure your database is in ARCHIVELOG mode. If not, enable it and provide the commands used. Configure the FRA if not set.
 - Connect to RMAN and display the default configuration using the appropriate command. Configure controlfile autobackup to ON if it's OFF. Include screenshots of the configuration output before and after changes.
- b) Perform Full Backup using RMAN
 - Execute a full database backup including archive logs. List the backups to verify. Provide the commands and a screenshot of the successful backup completion message.
- c) Simulate the database failure
 - Shut down the database cleanly.
 - Simulate a datafile loss by deleting one non-system datafile (e.g., the USERS tablespace datafile; identify its path via `SELECT name FROM v$datafile;`). Do not delete critical files like SYSTEM or SYSAUX in this

step to avoid full recovery simulation. Restart the instance and note the error (e.g., ORA-01157). Include a screenshot of the error message.

- d) Restore and Recover the Datafile.
- Connect to RMAN, take the affected datafile offline if needed (via SQL in RMAN: SQL 'ALTER DATABASE DATAFILE x OFFLINE';).
 - Restore and recover the specific datafile using RMAN commands (e.g., RESTORE DATAFILE x; RECOVER DATAFILE x;).
 - Bring the datafile online (SQL 'ALTER DATABASE DATAFILE x ONLINE';).
 - Provide commands and screenshots of the restore/recover success and verification query (e.g., SELECT status FROM v\$datafile WHERE file#=x; showing 'ONLINE').

[the x refers to the file number (or absolute file identifier) of the datafile that is affected]

- e) Full Database Recovery Simulation
- Shut down the database and simulate complete loss by deleting SPFILE, control files, and all datafiles (backup must exist).
 - Use RMAN to restore SPFILE, control file, database, recover, and open with resetlogs. Provide all commands in sequence and screenshots of key steps (e.g., restore completions).
- f) Validation and Cleanup
- Validate the entire database for corruption post-recovery.
 - Query a table to confirm data integrity. Include screenshots of validation output.

12. Learning about Standard Auditing in Oracle.

- a) Preparation and Configuration for standard auditing
- Connect as SYS and ensure the audit_trail parameter is set to DB (or DB, EXTENDED for more details). If not, set it and restart the database. Provide the commands and a screenshot of the parameter value from SHOW PARAMETER audit_trail;
- b) Enable auditing for all DML (INSERT, UPDATE, DELETE) and SELECT statements on the HR.EMPLOYEES table by access. Use the appropriate AUDIT commands. Include a screenshot of the success messages
- c) Generate Audit Records.
- Connect as the HR user and perform at least two auditable actions: one SELECT and one UPDATE (e.g., UPDATE EMPLOYEES SET salary = salary + 100 WHERE employee_id = 100; COMMIT;). Provide commands and screenshots of the execution outputs.
- d) Query and Analyze Audit Trail.
- Connect back as SYS and query DBA_AUDIT_TRAIL to view the generated records for the EMPLOYEES table. Filter by username='HR' and recent timestamps. Explain what each column (e.g., ACTION_NAME, SQL_TEXT if extended) represents. Include a screenshot of the query results showing at least two records.
- e) Disable audit policy and purge audit trial.

Disable the auditing policies set in step b). Purge the audit trail using DBMS_AUDIT_MGMT. Provide commands and screenshots of success messages.

f) Validation.

Perform another SELECT as HR and confirm no new audit record is generated by re-querying DBA_AUDIT_TRAIL. Include a screenshot of the empty or unchanged results post-disable.

13. Working with Unified Auditing in Oracle

a) Preparation and Verification.

Connect as SYS and verify unified auditing is enabled using the appropriate query on v\$option. If 'FALSE', note it (but proceed as mixed mode allows unified). Provide the command and a screenshot of the result.

b) Create and Enable Audit Policy

Create a unified audit policy to audit all DML (INSERT, UPDATE, DELETE) and SELECT on HR.EMPLOYEES. Enable the policy globally. Include commands and screenshots of success messages.

c) Generate Audit Records.

Connect as HR and perform at least two actions: one INSERT (e.g., INSERT INTO EMPLOYEES (employee_id, first_name, last_name, email, hire_date, job_id, salary) VALUES (999, 'Test', 'User', 'test@example.com', SYSDATE, 'IT_PROG', 5000); COMMIT;) and one DELETE (e.g., DELETE FROM EMPLOYEES WHERE employee_id = 999; COMMIT;). Provide commands and screenshots of executions.

d) Query and Analyze Unified Audit Trail.

Connect back as SYS and query UNIFIED_AUDIT_TRAIL for records related to EMPLOYEES, filtering by DBUSERNAME='HR'. Explain key columns like SQL_TEXT and EVENT_TIMESTAMP. Include a screenshot of results showing at least two records.

e) Conditional Auditing.

Modify or create a new policy to audit only when salary > 10000 (e.g., CREATE AUDIT POLICY hr_high_salary_pol ACTIONS UPDATE ON HR.EMPLOYEES WHEN 'NEW.SALARY > 10000' EVALUATE PER STATEMENT;). Enable it, perform a qualifying UPDATE, and query the trail. Provide commands and screenshots.

f) Disable, Drop, and Cleanup

Disable and drop all policies created. Purge the unified audit trail. Confirm no new records by performing an action and re-querying. Include commands and screenshots of empty results post-cleanup.